

STITCH: A Spatio-Temporal Integration Tool for Contextual and Historical enrichment of survey data.

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Software

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Summary

STITCH is a Python tool for linking high-resolution spatiotemporal datasets (e.g., decades of daily temperature records) to large-scale population survey data based on person-specific interview dates and geocoded residential identifiers at arbitrary spatial resolutions (e.g., census tract FIPS codes, ZIP codes). It supports user-defined temporal lag specifications (hourly, daily, or monthly) and accounts for participants' residential mobility, so linkage remains accurate across periods of residence and relocation while preserving high temporal and spatial specificity. Because STITCH runs entirely within a user's local computing environment, it is well suited to restricted-data settings in which participants' geographic identifiers cannot be externally released, enabling secure linkage without relying on inflexible, pre-linked de-identified datasets. By automating complex spatiotemporal joins and executing them fast enough to iterate on within a working session, STITCH lets researchers use high-dimensional contextual data that conventional merging approaches make difficult to integrate. It is distributed as a packaged executable with a graphical user interface (GUI) and a command-line interface (CLI), with the open Python source available for further customization.

Statement of need

The rapid expansion of geographically indexed, high-resolution spatiotemporal data through platforms such as Harvard Dataverse ([Harvard Dataverse, n.d.](#)), the Contextual Data Resource (USC/UCLA Center on Biodemography and Population Health, [n.d.](#)), ICPSR ([Inter-university Consortium for Political and Social Research, n.d.](#)), and the CDC/ATSDR Environmental Public Health Tracking Network ([Centers for Disease Control and Prevention, n.d.](#)) lets survey-measured outcomes be studied within environmental and historical context (e.g., linking individual health records to residential PM_{2.5} concentrations over time). Implementing such linkages, however, remains technically demanding when high temporal precision is required: exposure data must be aligned to person-specific interview dates and individual-level exposure windows constructed, including lagged and cumulative measures. Because interview timing varies across individuals, linkage cannot rely on simple merges. High-frequency data such as daily air pollution measures are therefore often collapsed into coarse summaries (e.g., annual means), which forecloses the study of immediate effects following time-sensitive exposures such as pollution spikes and natural disasters.

Longitudinal cohort studies add further complexity. Because participants may relocate between waves and data collection spans months, the geographic identifier used for linkage must be reassigned for each respondent at each wave and updated after each move. The challenge is compounded in restricted-data environments, common in population surveys because of confidentiality protections around residential information, where sharing is typically limited to pre-linked, de-identified datasets prepared inside secure enclaves. Such datasets are rarely

sufficient: different research questions require different contextual sources, temporal windows, and lag specifications.

STITCH addresses these barriers. It links spatiotemporal data to survey data on person-specific dates and geographic identifiers under user-defined lag specifications, accommodates mobility by reassigning linkage identifiers at each wave and after each move, and runs entirely within a user's secure local system. Its audience is survey-based researchers in epidemiology, gerontology, demography, and the social sciences who need exposure histories at the resolution at which the contextual data were actually collected. Source code and compiled executables are available at <https://github.com/njw0709/stitch>.

State of the field

In standard workflows, spatiotemporal data are linked to survey data with key-based merges. The programs most used in the social and health sciences rely on equality-based joins: Stata's `merge` (StataCorp, n.d.), base R's `merge()` (Merge Function - RDocumentation, n.d.), and `dplyr` joins (Wickham et al., 2026) all match exactly on specified identifiers, and none natively supports the non-equi joins required to construct multi-lag structures (e.g., linking values for each of many prior days). SAS (SAS Institute Inc., n.d.) supports non-equi joins through SQL, but applying them across large datasets and many joins demands careful memory management to remain tractable. Faced with these limits, researchers resort to ad hoc strategies — pre-aggregating high-frequency data, generating shifted date variables, or performing iterative merges — and writing a fast, reproducible pipeline this way is error-prone: a data-type mismatch in geographic IDs silently produces failed joins, loading large contextual files in full can exhaust memory, and hand-written multi-day lag logic invites off-by-one errors.

Interval and non-equi joins can be implemented in R's `data.table` (Barrett et al., 2026), but doing so requires advanced programming skills and multi-step data engineering. Closer in purpose, the Social Spatial Science Research Data Infrastructure (SoRa) (SoRa - Data Linking Service, n.d.) has advanced standardized, privacy-compliant spatial integration for population surveys, supporting flexible spatial matching (e.g., point-to-point or buffer-based linkage) and interoperability within secure research environments (Bensmann et al., 2020). Its focus, however, is spatial integration rather than fine-grained temporal alignment, and it does not address time-varying residence within a lag window.

We therefore built STITCH rather than contributing to these tools, because the gap is not a missing feature in an existing join engine but a missing layer above them. Rather than reimplementing the equality-based join logic Stata, R, and SAS already handle well, STITCH concentrates on what they do not support: automated alignment of high-frequency contextual data with time-varying participant locations and person-specific dates, under user-defined lag specifications, in a single reproducible, locally executable workflow.

78 Software design

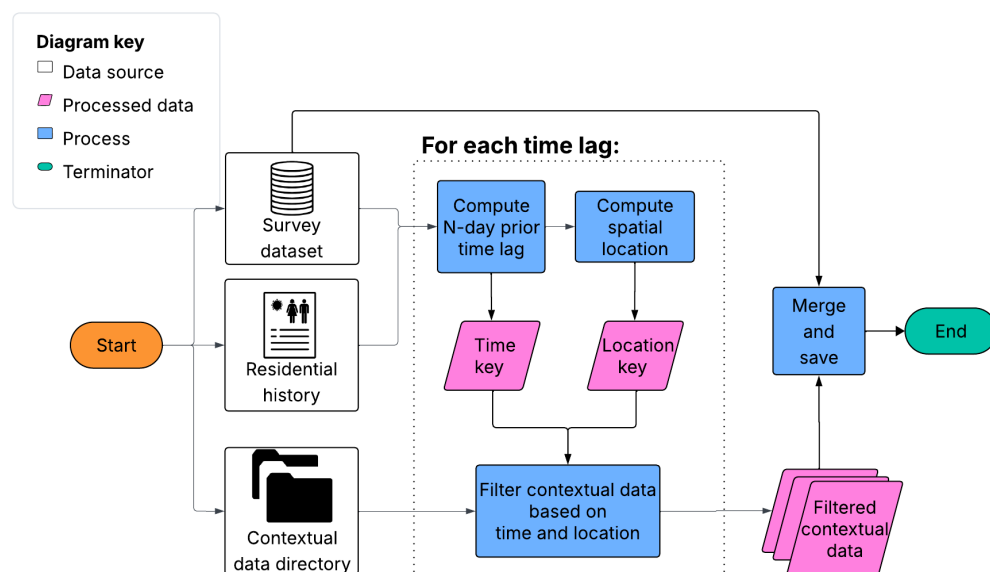


Figure 1: Overview of the STITCH processing pipeline. STITCH merges a survey dataset with spatiotemporal data based on time and location information derived from both the survey dataset and the residential history dataset.

STITCH takes three inputs (Figure 1): a survey dataset with participant ID, date, and geographic identifier (GEOID) columns; an optional long-format residential history, one row per residence; and a directory of long-format contextual data files, each carrying a period token in its filename. Formats, worked examples, and interface manuals are in the supplementary appendix (appendix.md) distributed with this paper.

Two interfaces over one core, with the GUI as a working tutorial. The GUI and CLI are thin shells over a single configuration object and one linkage implementation, so a researcher's technical background determines only ergonomics, never capability: a job configured in the packaged executable performs the identical linkage as the equivalent command-line invocation, and the open Python source underneath remains available to users who want to customize the pipeline further. We treat the GUI as more than an accessibility shim. Its wizard steps deliberately mirror the pipeline stages in Figure 1 — select the survey data and name its ID, date, and GEOID columns; optionally attach a residential history; point at the contextual data directory; set resolution and lag window; run — so a newcomer who has never performed a spatiotemporal linkage learns what the pipeline requires, and what it does, by being walked through it. That structure is also where the checks live that make otherwise silent failure modes visible. The failure most likely to corrupt results is a GEOID mismatch — leading zeros lost when a FIPS code is read as a number — which produces empty joins rather than errors, so identifier normalization is an explicit, previewable step: users treat identifiers as codes (strings, optionally zero-padded to a fixed width) or as numeric values, and see the transformation before running. In the same spirit, every step reports what it found and inferred — which contextual files matched, a preview of one, the temporal resolution read from them, how sampled move dates were parsed — column roles are validated as they are assigned, and lag requests finer than the contextual data's own resolution are refused rather than silently upsampled. Because CLI flags correspond one-to-one to the wizard fields, this understanding transfers directly to scripted use: the GUI is documentation that executes, and it stays useful to users who ultimately work from the command line. The GUI additionally maintains a job

queue, so several runs (different measures, lag windows, or outputs) can be configured and then executed sequentially under one progress dialog with a live log.

Lags as precomputed keys, not as a non-equi join. The central design decision is to avoid non-equi joins entirely. Instead of asking a join engine to match rows within a temporal interval, STITCH materializes, for each requested lag, the (lag date, GEOID) key that each respondent implies, and then performs an ordinary equality join against the contextual data. This converts one hard join into k easy, mutually independent ones. The cost is that the lag structure is fixed before the merge rather than expressed declaratively; the benefit is that every step is vectorized, memory-bounded, and embarrassingly parallel. It is also what makes the mobility logic tractable, since the GEOID is simply part of the key being computed.

Speed as a design goal, not an afterthought. That structure is what makes STITCH fast, and speed is what makes flexible linkage practical rather than merely possible. On a desktop CPU and without parallelism, STITCH links a full year of daily contextual data — 10,000 respondents by 365 lags — in 9.7 seconds. Runtime scales sub-linearly with survey size (a 50-fold increase in rows raised runtime about 14-fold) and improves per unit of work as the window grows, per-lag cost falling from 64.2 ms at 7 lags to 26.6 ms at 365 as the fixed cost of reading and filtering contextual data is amortized; a 20-fold increase in unique GEOIDs raised runtime by only 54%. Parallel lag processing roughly halves total runtime, the worker count capped automatically from available memory so the speedup cannot push the machine into swapping (full benchmarks in the appendix). Linkage therefore becomes something a researcher iterates on within a session — a different lag window, measure, or resolution — rather than an overnight batch job whose cost discourages revision, which is what makes the aggregation shortcut avoidable: keeping daily resolution no longer carries a prohibitive price.

Resolve location before merging. Residential mobility gets a stage of its own: STITCH resolves each person-period pair to a GEOID before any contextual data are touched, rather than correcting locations after the fact. This isolates the survey-specific complexity in one auditable step that covers the scenarios real cohorts present without asking users to pre-flag them — multiple moves (the GEOID in effect at each lagged point, so values may differ across lags for one respondent), a single residence (propagated to every lag without extra rows), and left-truncation or right-censoring (missing values for the affected points, the rest linking normally, and a stitch-period average produced only for respondents linkable across the full window). Coarse dates are anchored to the midpoint of the period they denote rather than its start, minimizing worst-case dating error when surveys record only the month or year of an interview or a move.

Stream the large side; keep the small side in memory. Contextual archives routinely exceed available RAM, so STITCH inverts the usual load-then-filter order: it derives the required dates and GEOIDs from the survey and residential history first, then reads only the matching portions of the contextual files, writing each lag's result to an intermediate file and recombining at the end. Peak memory is governed by the survey data and one lag's slice rather than the archive's size, at the cost of extra disk I/O — a trade we accept because researchers' binding constraint is memory, not disk.

Local-first, with confidentiality as a hard constraint. The pipeline never requires geographic identifiers to leave the user's machine. This trades the scalability of centralized or cloud processing for the confidentiality guarantees restricted-use survey data demand, and follows through to details: intermediate per-lag files default to a hidden temporary location outside the output directory and are deleted on successful completion, so confidential partial products are not left behind.

Research impact statement

STITCH's significance lies in being reusable methodological infrastructure rather than one study's analysis code, and its use to date is the evidence for that claim. It was built to link

156 heat exposure to epigenetic aging measures in the Health and Retirement Study (E. Y. Choi
157 & Ailshire, 2025; University of Michigan, 2025), but the same pipeline — re-specified rather
158 than rewritten — has since supported studies of air pollution (Van Dang, Choi, Crimmins,
159 Finch, et al., 2025), kidney function (Van Dang, Choi, Crimmins, & Ailshire, 2025), adaptive
160 immunity (E. Choi et al., 2024), and machine learning analyses of population heterogeneity
161 in environmental health risk (Nam et al., 2026). These span different exposures, outcomes,
162 temporal windows, and analytic designs, and several were carried out by collaborators outside
163 the original development team, who identified issues and contributed feature enhancements.
164 The software has thus been hardened by use rather than by anticipation, and that one tool
165 spans this range of requirements — extended by its users rather than forked by them — is
166 what distinguishes durable infrastructure from a bespoke script.

167 Nothing in the design is specific to a single exposure or cohort: the linkage keys are simply
168 dates and geographic identifiers. The framework therefore generalizes to any dataset carrying
169 time and location information that must be aligned to external spatiotemporal sources, and
170 as high-resolution exposure assessment becomes central to epidemiologic and demographic
171 research, STITCH is positioned for reuse and citation well beyond the studies above.

172 AI usage disclosure

173 Portions of the STITCH codebase, including specific unit tests, were developed with the
174 assistance of large language models (LLMs). The authors reviewed and verified all AI-assisted
175 contributions.

176 License

177 STITCH is released under the MIT License.

178 Author contributions

179 Jennifer A. Ailshire, Eun Young Choi, and Jong Woo Nam conceptualized STITCH and
180 designed its core functionality. Jong Woo Nam developed and implemented the software. Eun
181 Young Choi and Jong Woo Nam conducted comprehensive functionality testing across multiple
182 use-case scenarios, with primary testing performed on Windows and Linux platforms.

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